

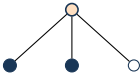
Definition and first terms

$4\,A(x)=3+1\,x+A(x)^3$, with the origin-normalized solution.
 Normalization/grammar: $A(x)=1+(1)*T(x)$; $T=x+3*T^2+1*T^3$.
 $a(0),\ldots,a(7)=1,1,3,19,150,1326,12558,124590$.
 Kernel used throughout: $\rho(u)=u\,(1-3\,u^1-1\,u^2)$, so $x=\rho(T(x))$.

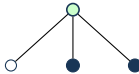
Typogeometry



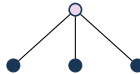
1x1



{1,1,0}x1



{0,1,1}x1



{1,1,1}x1

literal 3-slot geometry; empty positions are shown. Filled endpoints are true leaves; open endpoints are false/empty leaves. For colored grammars, color is genuine constructor data and is not silently converted into a positional zero pattern.

Typogeometric constructor codes and multiplicities: $\Delta_2\colon 3$, $\Delta_3\colon 1$.

Coefficient and contour extraction

$$\mathcal{K}_n=\{(k_1,\ldots,k_2)\in\mathbb{Z}_{\geq 0}^2:\sum_{j=1}^2jk_j=n-1\},\quad K=\sum_{j=1}^2k_j,$$

$$a(n)=\frac{1}{n}\sum_{\mathbf{k}\in\mathcal{K}_n}\binom{n+K-1}{K}\binom{K}{k_1,\ldots,k_2}3^{k_1}1^{k_2}.$$

$$a(n)=\frac{1}{2\pi i n}\oint_\gamma\frac{du}{\rho(u)^n},\quad n\geq 1.$$

Recurrence

$$\sum_{r=0}^2P_r(n)a(n+r)=0,\,n\geq 1,\,\text{with}$$

$$\begin{aligned}P_0(n)&=-27\,n^2+3\\P_1(n)&=-162\,n^2-243\,n-81\\P_2(n)&=13\,n^2+39\,n+26\end{aligned}$$

Differential equation

$$(3\,x^2)\,A(x)+(-27\,x^3-81\,x^2)\,A'(x)+(-27\,x^4-162\,x^3+13\,x^2)\,A^{(2)}(x)=0.$$

Matrices, telescoper certificate, and checks

$$U=\begin{pmatrix}\frac{153}{13}&\frac{24}{13}&\frac{3}{13}\\\frac{72}{13}&\frac{9}{13}&\frac{6}{13}\\0&0&0\end{pmatrix},\quad V=\begin{pmatrix}-1&0&0\\\frac{75}{13}&\frac{11}{13}&\frac{3}{13}\\\frac{24}{13}&\frac{3}{13}&\frac{2}{13}\end{pmatrix},\quad J=\begin{pmatrix}0&1&0\\0&0&2\\0&0&0\end{pmatrix}$$

Certificate.

$$R(n,u)=\frac{-9\,n\,u^4-36\,n\,u^3+6\,n\,u^2+84\,n\,u-13\,n-3\,u^4-12\,u^3-6\,u^2+3\,u}{\rho(u)^1}.$$

Telescoping identity: $\sum_rP_r(n)H_{n+r}(u)=\partial_u(R(n,u)H_n(u))$.

Matrix dimensions: 6×6 ; remainder 2×3 , rank 2, nullity 1. The exact telescoping residual is zero. The recurrence reconstructs all 24 stored terms; 6/6 published OEIS prefix terms match.